XRPL Validator — Daily Engineering Report · 2026-10-15

Generated 2026-10-16 00:21 EDT. Scope: XRPL work only — the git history of xrpl-validator and the XRPL Rust SDK, the live native-engine shadow, the reference rippled node, the XRP Ledger itself (amendments) and XRPL research.

Summary

- **12 commits** (xrpl-validator 12)
- Live shadow: 10,937/10,937 ledgers fully matched, 1,236,407 transactions, 0 engine receipts in the current soak
- Amendment **fixCleanup3_4_0** held majority (since 2026-10-09 09:34 UTC); activation due ~2026-10-23 09:34 UTC

Live validator and native engine

Snapshot at generation time.

Measure	Value
Ledgers compared (current run)	10,937
Ledgers fully matched	10,937
Transactions applied natively	1,236,407
Result mismatches	0
Byte mismatches	0
Missing / extra keys	0 / 0
Batch inner-result mismatches	0
Engine receipts (current soak)	0
Live engine build running since	2026-10-15 12:29 EDT
Reference rippled	3.4.1, full, validated #107,687,396

Engineering changes

- **feat(peer): Independence 7.1 — answer regular peers' pings (XRPL_PEER_PONG=1, off by default); the plan gains steps 7-9** (`6d81d3e`, xrpl-validator) Regular peer sessions never answered the 60-second keep-alive ping, so every server dropped the validator host after about two minutes and it dialled again: about 25,000 handshakes in the first 10 hours of cycle 176, for some 80 peers. Only the 4 relay sessions answered (peer::relay). With XRPL_PEER_PONG=1 every session answers with peer::codec::pong_frame_for (tested in codec.rs), before the duplicate filter. /api/peers gains …
- **fix(peer): 7.1's pong is written from its own task, so a GetLedger reply holding the writer through a long write to a slow peer never stalls the read loop (a5's review)** (`0c32cdf`, xrpl-validator)
- **feat(peer): Independence 7.2-7.4 — one node key, private by design, checked transaction relay (all off by default)** (`5a2c738`, xrpl-validator) 7.2 XRPL_NODE_IDENTITY=1: every regular peer session shows one node

key, loaded from paths::node_seed_path (XRPL_NODE_SEED_PATH, default ${XRPL_DATA_DIR}/node_seed.hex, 0600, created on first use by load_or_create_seed), never the validator's seed (refused if the paths match). Without it each connection made a fresh key, so every server met a stranger each time the validator host dialled it. A seed that can't be used falls back …

- **feat(validator): Independence Step 9 — clock check, pending-only amendment votes, the list refreshed while running (all off by default)** (`c8c361d`, xrpl-validator) Clock (XRPL_CLOCK_CHECK=1, XRPL_CLOCK_GATE=1): clock_check.rs, a small SNTP client (RFC 4330, no new dependency) that asks time.google.com, time.cloudflare.com and pool.ntp.org every 5 minutes and keeps the median offset: a log line when it changes, `clock` in /api/engine. rippled 3.x has no such check (TimeKeeper::now() is the system clock, TimeKeeper.h:66). The gate skips signing while 2 or more servers agree the …
- **fix(unl): drop the unused read_vl_length import (clippy, a5's o183)** (`d7f5d8b`, xrpl-validator)
- **feat(peer-state): the plan records Step 8 built in shadow, with the measured full run** (`88df3b3`, xrpl-validator) docs/INDEPENDENCE-PLAN.md, Step 8: on .220, ledger #107674282 (a 28/28 quorum of the verified list), all 20,031,916 objects from 12 mainnet peers in 14 minutes (5,436 requests, 7.0 GB in, 0.2 GB of memory, no timeout, refusal or disconnect); the root rebuilt bottom-up from the file equals the account_hash, and the reference node agreed as a witness. What the source switch still needs: writing straight into state.rocks and using …
- **feat(peer-state): per-peer reply sizes, latency, timeouts and close reasons at the end; discovery in the progress line** (`79e684c`, xrpl-validator) What each peer allowed is the point of a shadow run: the biggest reply (nodes and MB), the average time to answer, the timeouts and why each session closed, printed per peer after the run, and the addresses TMEndpoints brought in the progress line and the summary.
- **feat(peer-state): Independence 8 in shadow — a ledger's whole state tree from mainnet peers, every node proven (peer_state_sync)** (`58158db`, xrpl-validator) A cold start downloads the state from the reference node's RPC (ledger_data, bulk_sync.rs); nothing fetched it from peers. peer_state does what rippled's InboundLedger does: TMGetLedger liAS_NODE by position, each node taken only where a proven parent named it and only when it hashes to that name (inner: MIN over the 16 children; leaf: MLN over the object and its key, the key under its position), starting from the header's …
- **fix(peer-state): depth.is_multiple_of(2) (clippy, a5's o184)** (`046efb7`, xrpl-validator)

Docs and operations

- **chore(tools): xrpl-watch moved to its own repo (ilubxrp589/xrpl-watch); this folder keeps a pointer** (`1711cca`, xrpl-validator)
- **chore: scrub a login password and home-network addresses from docs, the dashboard config and a script** (`0bb7783`, xrpl-validator)
- **chore(node): the source names no home-network address** (`67ce8ff`, xrpl-validator)

Amendment watch

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

Amendment	Majority since	Activation due
fixCleanup3_4_0	2026-10-09 09:34 UTC	2026-10-23 09:34 UTC

Amendments enabled on mainnet: 97.

Appendix — commits

Time	Repo	Commit	Subject
09:30	xrpl-validator	`6d81d3e`	feat(peer): Independence 7.1 — answer regular peers' pings (XRPL_PEER_PONG=1, off by default); the plan gains steps 7-9
09:31	xrpl-validator	`0c32cdf`	fix(peer): 7.1's pong is written from its own task, so a GetLedger reply holding the writer through a long write to a slow peer never stalls the read loop (a5's review)
09:49	xrpl-validator	`5a2c738`	feat(peer): Independence 7.2-7.4 — one node key, private by design, checked transaction relay (all off by default)
10:03	xrpl-validator	`c8c361d`	feat(validator): Independence Step 9 — clock check, pending-only amendment votes, the list refreshed while running (all off by default)
10:05	xrpl-validator	`d7f5d8b`	fix(unl): drop the unused read_vl_length import (clippy, a5's o183)
10:29	xrpl-validator	`88df3b3`	feat(peer-state): the plan records Step 8 built in shadow, with the measured full run
10:29	xrpl-validator	`79e684c`	feat(peer-state): per-peer reply sizes, latency, timeouts and close reasons at the end; discovery in the progress line
10:29	xrpl-validator	`58158db`	feat(peer-state): Independence 8 in shadow — a ledger's whole state tree from mainnet peers, every node proven (peer_state_sync)
11:45	xrpl-validator	`046efb7`	fix(peer-state): depth.is_multiple_of(2) (clippy, a5's o184)
14:54	xrpl-validator	`1711cca`	chore(tools): xrpl-watch moved to its own repo (ilubxrp589/xrpl-watch); this folder keeps a pointer
16:22	xrpl-validator	`0bb7783`	chore: scrub a login password and home-network addresses from docs, the dashboard config and a script
16:36	xrpl-validator	`67ce8ff`	chore(node): the source names no home-network address